

Immanuel Peter

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EDUCATION

The University of Chicago – BS Computer Science, BA Physics | *expected June 2028*

Courses: Machine Learning, Abstract Linear Algebra, Systems Programming, Theory of Algorithms, Foundations of Distributed Systems

EXPERIENCE

Member of Technical Staff Intern | Tensormesh | *March 2026 – Present*

- Integrated Phoenix into Tensormesh's observability stack, turning router traffic into OpenInference traces and spans.
- Shipped SDK and CLI surfaces for inspecting traces/spans, including async clients, docs, generated references, and tests.
- Productionized rollout configuration across Terraform, Secret Manager, and Cloud Run
- Contributed vLLM Production Stack fixes for runtime defaults and router volume customization.

Software Engineer Intern | Quantum Rings | *June 2025 – August 2025*

- Reduced API request latency by designing queue-driven telemetry processing that decoupled heavy operations from the API, enabling horizontal scaling (AWS SQS, TypeORM)
- Built full-stack admin analytics dashboards tracking user growth and execution volume with SQL time-bucket aggregation and time zone safe filtering (Next.js, NestJS, Recharts)
- Delivered 19 PRs, 43 contributions, ~15K LOC added across schema migrations, background workers, and KPI dashboards

Open Source Contributor | Meta | *October 2025*

- Contributed to Pyrefly, Meta's Python type checker in Rust, refactoring error summarization module structures

PROJECTS

AutoMoE — MoE Self-Driving Model | github.com/immanuel-peter/self-driving-model

- Built a 53M parameter modular self-driving stack in PyTorch with 4 expert networks, top-2 gating, and 10-waypoint trajectory prediction.
- Released ~145K CARLA frames across multimodal and multi-camera datasets, and trained/fine-tuned the pipeline with PyTorch DDP on up to 8 GPUs.

CARLA Autopilot Datasets | hf.co/immanuelpeter/datasets | **15K+ Downloads**

- Released two MIT-licensed CARLA autopilot datasets (~145K frames, 567 GB) on Hugging Face with 2.2K and 6.5K downloads each; flagship split includes synchronized 4-camera RGB, 32-channel LiDAR, semantic segmentation, and 2D bounding boxes across 24 weather/map scenarios
- Designed run-based train/val/test splits to eliminate temporal leakage, supporting perception, sensor fusion, imitation learning, and RL benchmarking under extreme conditions

Hostess — Docker Compose for Production | hostess.sh

- Designed and built a multi-cloud PaaS control plane (Go CLI/API + Kubernetes controller + Next.js Studio) that compiles a declarative config into production workloads
- Implemented the deploy pipeline end-to-end: multi-backend image builds, dependency-ordered manifest generation, secret injection, readiness waits, and stable URL routing
- Built a connection-secret controller and dual-proxy database edge (Postgres SNI + Redis L7 routing) so managed DBs get correct credentials and secure external access without exposing cluster networking
- Shipped GitHub App CD with deploy-on-push, isolated per-PR preview environments, commit statuses, plus Studio surfaces for route metrics/traces, FastAPI OpenAPI playground, Next.js artifact introspection, and Postgres/Redis admin with backup/restore and PITR

Redis Kubernetes Operator | github.com/howl-cloud/redis-operator

- Engineered a production-grade Kubernetes operator in Go managing Redis standalone, sentinel, and cluster modes with direct pod orchestration for deterministic failover
- Achieved 606 reconciles/sec by implementing fencing-first failover with split-brain prevention, boot-time role guards, and supervised primary updates
- Built chaos test suite (network partition, primary kill, operator restart), Prometheus metrics, Grafana dashboards, and Helm chart with OCI registry publishing

Matchbox — AI Research Matching

- Built an AI-driven research lab matchmaking platform connecting students with labs using vector search and LLM-based scoring for semantic matching (Next.js, FastAPI, ChromaDB, OpenAI)
- Architected scalable backend infrastructure on GCP using Docker, Terraform, and Cloud Run with CI/CD pipelines for automated deployment and rapid iteration